

Time Series: Introduction

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Introduction

A time series is a sequence of observations indexed by time. Time-series analysis decomposes the series into four interpretable components — trend, seasonality, cyclical, and noise — and models each. The decomposition framework underpins virtually every classical method: ARIMA models the noise after differencing out trend and seasonality; ETS captures all four components in a state-space form; Bayesian structural time series isolates them as latent processes. Understanding the four-component vocabulary is the foundational step before reaching for any specific model.

Prerequisites

A working understanding of basic statistics, the `ts` class in R, and the difference between cross-sectional and longitudinal data structures.

Theory

The classical additive decomposition is

$$y_t = T_t + S_t + C_t + \varepsilon_t,$$

with components: trend T (long-term level change), seasonal S (regular fixed-period variation, e.g. monthly cycle), cycle C (irregular longer-than-seasonal swings, e.g. business cycles), and noise ε (residual stochastic variation). The multiplicative form $y_t = T_t \cdot S_t \cdot C_t \cdot \varepsilon_t$ applies when seasonal amplitude grows with the level; it is additive on the log scale.

In practice, the cycle is often combined with the trend (giving “trend-cycle” as one component), leaving three components in most decomposition implementations: trend-cycle, seasonal, and remainder.

Assumptions

Time order is preserved (no shuffling); regular sampling intervals (irregular sampling requires `zoo` / `xts` or modelling the spacing); the chosen decomposition (additive vs multiplicative) matches the data.

R Implementation

```
library(forecast)

data(AirPassengers)
plot(AirPassengers, main = "Monthly airline passengers")

# Classical decomposition
dec <- decompose(AirPassengers, type = "multiplicative")
plot(dec)

# STL: loess-based decomposition
```

```
stl_dec <- stl(log(AirPassengers), s.window = "periodic")
plot(stl_dec)
```

Output & Results

`decompose()` returns the classical additive or multiplicative decomposition; `stl()` returns the loess-based seasonal-trend decomposition. The four-panel plot shows observed, trend, seasonal, and remainder components stacked vertically.

Interpretation

A reporting sentence: “The monthly air-passenger series exhibited a clear upward trend (approximately 5 % per year), a stable annual seasonal cycle peaking in summer, and noise that grew with the level (motivating the log transformation). STL decomposition on the log scale produced trend, seasonal, and approximately white-noise remainder components.” When introducing a time series, describe each component visible in the data.

Practical Tips

- Log-transform series whose variance grows with the level; the resulting decomposition on the log scale is additive and well-behaved.
- STL (`stl()`) is more robust than classical `decompose()`; the latter is sensitive to outliers and assumes fixed seasonal patterns.
- After decomposition, examine the remainder series with ACF / PACF plots and Ljung-Box; substantial autocorrelation indicates missed structure.
- The modern tidy workflow uses `tsibble` for time-aware tibbles, `feasts` for decomposition and diagnostics, and `fable` for forecasting; together they replace much of the older `forecast` workflow.
- For irregular sampling intervals, use `zoo` or `xts` time series; classical decomposition assumes regular sampling.
- For multiple seasonalities (e.g. weekly + yearly in retail data), `forecast::mstl()` produces a multi-layer decomposition that single-seasonal STL cannot.

R Packages Used

Base R `ts()`, `decompose()`, and `stl()`; `forecast` for the broader classical workflow including `auto.arima`, `forecast`, `mstl`; `tsibble`, `feasts`, `fable` for the modern tidyverts ecosystem; `zoo` and `xts` for irregularly-spaced time series.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics